

Krish Shrivastav

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PROFESSIONAL SUMMARY

Third-year BCA student who builds systems end-to-end: a key-value database engine that survives a crash without losing a single write, and a real-time canvas where multiple people draw together with zero merge conflicts. Comfortable across C++, Python, JavaScript, React, Node.js, PostgreSQL/Oracle, REST APIs, and Git.

EDUCATION

Bachelor of Computer Applications (BCA) | 2024 – 2027
Sinhgad College of Science, Savitribai Phule Pune University (SPPU)
SGPA: 9.36 / 10

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript
Web: HTML, CSS, React.js, Node.js, Flask
Databases: SQL, PostgreSQL, Oracle
Tools & Concepts: Git, GitHub, WebSockets, REST APIs

PROJECTS

Collab Whiteboard – Real-Time Collaborative Canvas

[Live Demo](#) | [GitHub](#)

- Enabled multiple users to draw on the same canvas at once, seeing each other's strokes and live cursor positions in real time — built with React, Konva.js, Node.js, and WebSockets.
- Resolved simultaneous edits automatically with Yjs CRDTs, so concurrent strokes from different users merge without conflicts or lost work; cursor positions stream on a separate ephemeral channel to keep drawing responsive.
- Gave each user independent undo history and persisted every shape to disk, so a session survives a refresh or reconnect.

MiniDB – Key-Value Database Engine

[Live Demo](#) | [GitHub](#)

- Built a key-value database engine from scratch in Python — no third-party dependencies in the core — that survives a crash without losing data: every write lands in an append-only log with CRC32 checksums, replayed on restart to rebuild state.
- Answered point lookups in $O(1)$ via an in-memory primary index, and range/equality queries in $O(\log n)$ via a self-built, self-balancing B+ tree for secondary indexes.
- Let keys expire automatically via TTL, grouped writes into atomic transactions, and kept concurrent access safe with thread-level and process-level (file) locking, with background compaction keeping the log from growing unbounded.
- Exposed the engine over both a raw TCP protocol and a Flask HTTP API, so it's usable as an embedded store or a networked service.

ACTIVITIES & INTERESTS

Recreational sports: cricket, football, badminton. Casual gaming: Valorant, Mobile Legends: Bang Bang (MLBB), BGMI, CS:GO.

LANGUAGES

Hindi, English, Marathi